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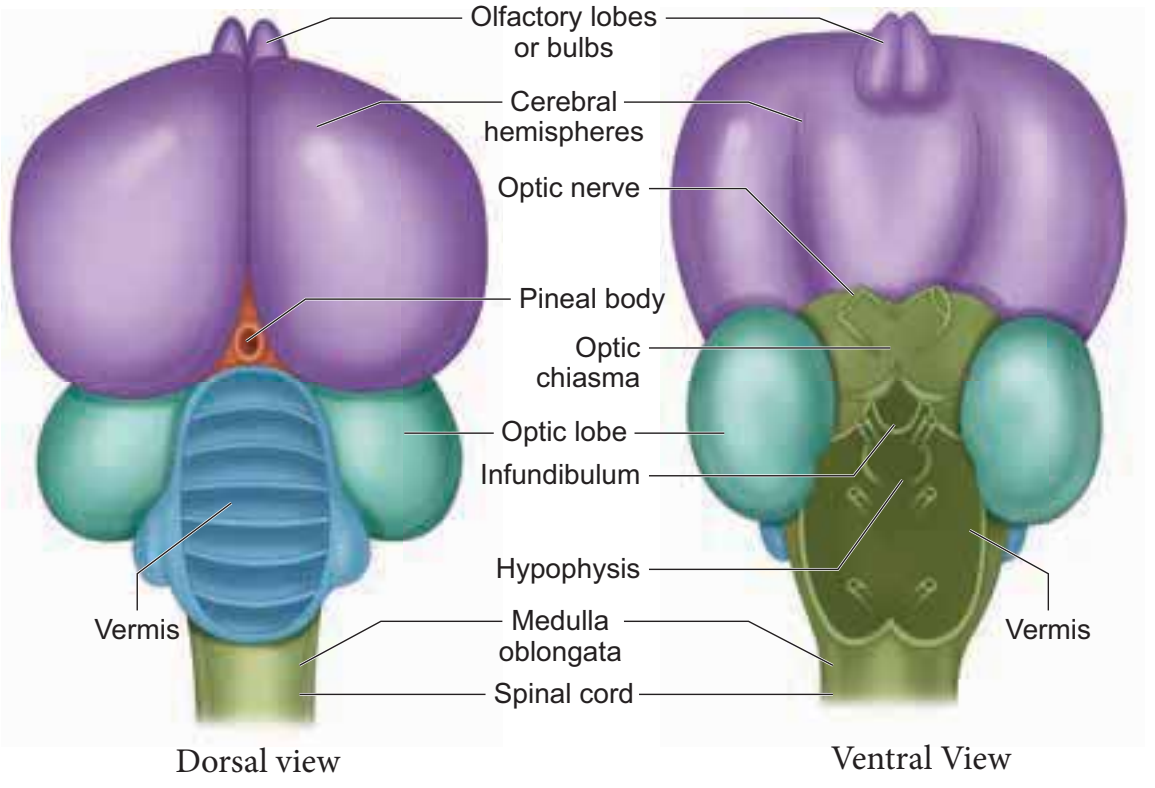

Figure 4.34 Pigeon - Structure of brain

This vein runs forward, receives in its course veins from the rectum, and joins the **hepatic portal vein.** The **epigastric vein** returns the blood from the mesenteries and joins one of the hepatic veins.

Nervous system and receptor organs

The nervous system consists of **central nervous system** which includes the brain and spinal cord, the **peripheral nervous system** and the **autonomous nervous system** (Figure 4.34). The brain of pigeon is larger than in lower forms, it is short, broad and rounded within cranial cavity. It is covered by two **meninges**, the outer **duramater** and an inner **pia- arachnoid** membrane and the space between the two meninges is filled with **cerebrospinal fluid**. The **cerebral hemispheres** of the pigeon are large and extend behind to meet the **cerebellum**. The cerebrum controls voluntary movements and is the centre for memory and intelligence. The **diencephalon** is covered dorsally by the cerebral hemispheres and cerebellum. The diencephalon relays impulses to the cerebral hemispheres, integrates the **autonomic system** and the perception of extreme cold, pain, heat etc. On the ventral side of the diencephalon is the **optic chiasma,** behind the chiasma projects the **infundibulum** bearing a large **hypophysis** or **pituitary**. The optic lobes are large and occupy a lateral position owing to the large size of the cerebral hemispheres and cerebellum. Optic lobes are centres for sight. The pineal body and infundibulum are present. The cerebellum is highly developed and convoluted indicating the delicate sense of equilibrium and the great power of muscular co-ordination required for birds.